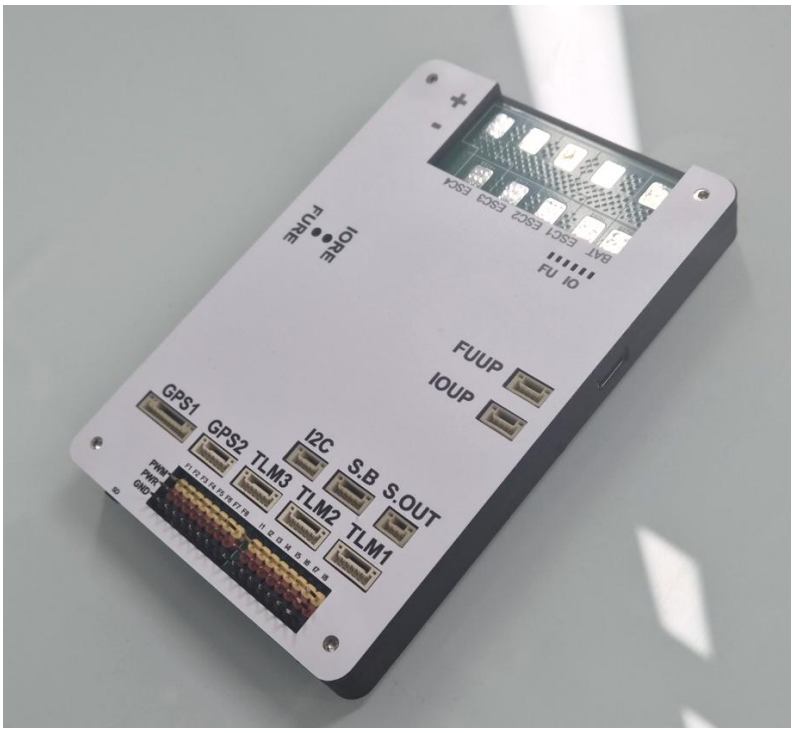
Corp.DCLASS

DCD FC

Integrated power distribution flight controller | Product and connection guide

DCD FC is a flight controller with integrated power distribution developed by Corp.DCLASS. It combines battery input and power connections for four external ESCs on one board. GPS, telemetry, receiver interfaces and 16 PWM outputs support aircraft integration.



DCD FC with enclosure

Specifications

Processors	STM32H743IIT6 main MCU / STM32F103C8T6 IO MCU
IMUs	BMI088 + ICM-42688-P accelerometers and gyroscopes
Barometer / compass	MS5611 / IST8310
Storage	FRAM parameters / microSD flight logs
Outputs / interfaces	16 PWM outputs, 2 GPS, 3 telemetry, I2C, SBUS input, USB
Input voltage	Up to 25.2 V (6S LiPo) / supports 3–6S
Power distribution	Power pads for 4 external ESCs / ESCs not included
Size / weight	Approx. 126 × 84 × 13 mm / 103 g with enclosure

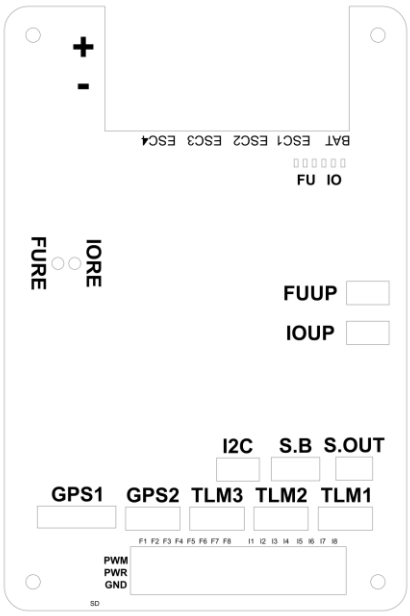
Dimensions refer to the enclosure, excluding protruding connectors and pins.

Purchase and technical support | Corp.DCLASS | https://www.d-class.co.kr

Connect to a PC using a standard USB data cable. Flight logs are stored on microSD.

Ports and connections

Use the top-panel labels and the table below to identify ports and their functions.



BAT / ESC1~4	Battery input / external ESC power
GPS1 / GPS2	GPS / SERIAL3 / SERIAL4
TLM1 / TLM2	Telemetry / SERIAL1 / SERIAL2
TLM3	Additional UART / SERIAL5
I2C	External compass and I2C peripherals
S.B	SBUS receiver input
F1~F8	FMU PWM / SERVO9–16
I1~I8	IO PWM / SERVO1–8
USB / SD	PC connection / microSD logs
FUUP / IOUP	FMU / IO programming via SWD

S.OUT is the IO SBUS output. See page 8 for FURE/IORE reset buttons and FU/IO LEDs. PPM input and external CAN ports are not supported.

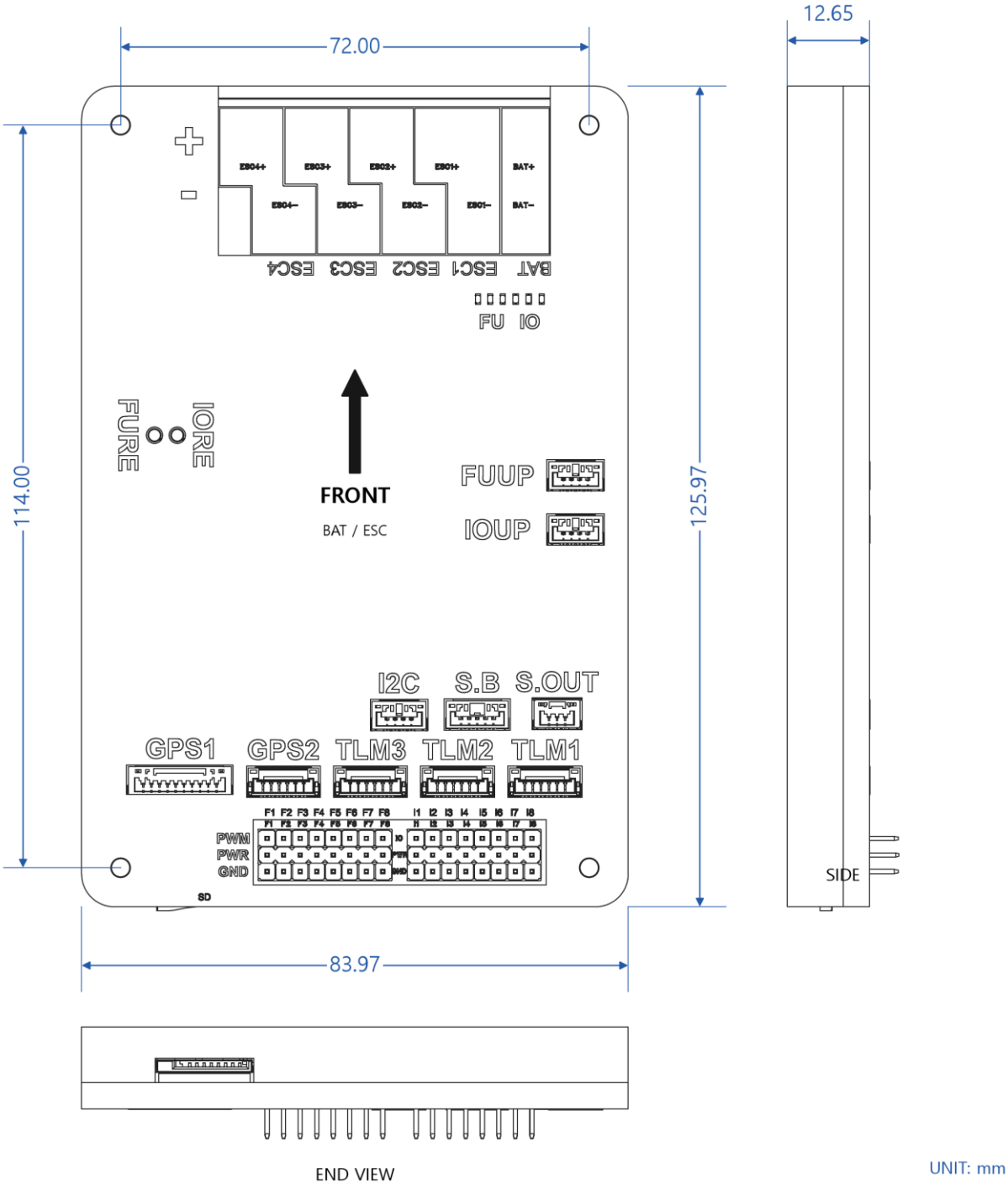
Port power supply

Port / power pin	Voltage	Supply current and sharing
GPS1·GPS2 / TLM1~3 / I2C / S.B	5 V	Up to 1.5 A total, including internal FC consumption.
F1–F8 + I1–I8 / PWR row	5 V	Up to 1.5 A total, shared by F1–F8 and I1–I8. Separate from the FC/peripheral supply.
GPS1 / pin 3	3.3 V	1 A supply rating, shared with microSD.
S.OUT	—	No power output pin.
FUUP / IOUP / pin 1	3.3 V REF	Debugger voltage reference; do not use to power external devices.
BAT / ESC1~4	Battery voltage	Battery distribution to external ESCs. Up to 25.2 V (6S LiPo).

Current figures apply to each shared supply as a whole. Allow for internal consumption and thermal limits. USB power is also subject to host and cable limits.

Dimensions and forward direction

Front points toward the BAT and ESC power pads.



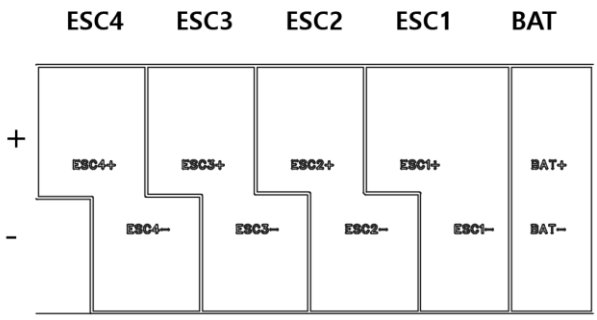
UNIT: mm

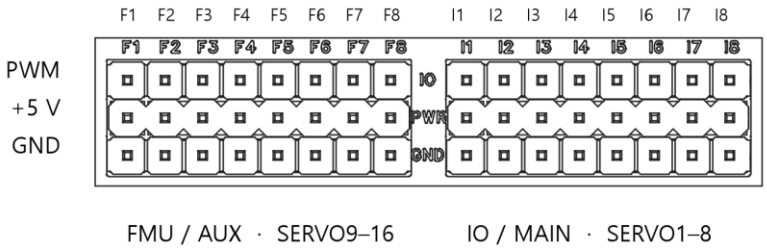
Dimensions in mm refer to the enclosure, excluding protruding connectors and pins. The 72 × 114 mm hole pattern is for enclosure assembly.

Secure the FC to the aircraft with double-sided tape. Allow clearance for USB, microSD, connectors and protruding pins.

Power connections and PWM outputs

Top view with BAT/ESC at the top. BAT is the battery input; ESC1–4 are power pads for external ESCs.





IO PWM OUT = MAIN OUT	I1~I8 → SERVO1~8
FMU PWM OUT = AUX OUT	F1~F8 → SERVO9~16

The PWM PWR row supplies 5 V shared by IO and FMU outputs. Do not connect another BEC in parallel. Input supports 3–6S LiPo, up to 25.2 V (6S).

Battery monitor settings

Mission Planner → CONFIG → Full Parameter List

Voltage multiplier	BATT_VOLT_MULT	8.987575
Current scale (optional)	BATT_AMP_PERVLT	0.25 A/V
Current offset	BATT_AMP_OFFSET	0 V

BATT_MONITOR = 3 (default: voltage only) | BATT_VOLT_PIN = 8

For optional FC current monitoring, set BATT_MONITOR=4 and BATT_CURR_PIN=4, then reboot. Adjust voltage/current scale parameters against a reference meter.

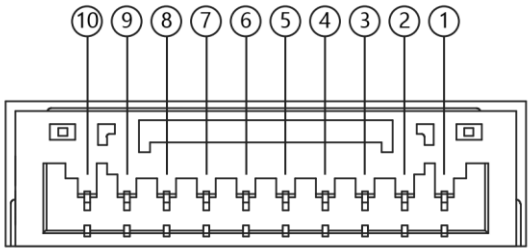
The sensor measures only the FC supply branch, excluding ESC current. Use an external sensor for total battery current, consumption and remaining capacity.

GPS ports

Top view with Front at the top. Circled numbers identify pins. Left and right are reversed when viewed from the cable plug mating face.

GPS1

SERIAL3 / USART1

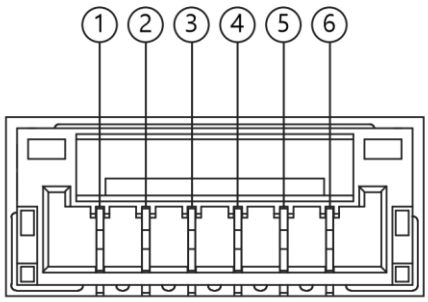


10 pins / 1.25 mm pitch

Pin	Signal
1	GND
2	BUZZER
3	+3.3 V
4	SAFETY LED
5	SAFETY SW
6	SDA
7	SCL
8	RX
9	TX
10	+5 V

GPS2

SERIAL4 / UART8



6 pins / 1.25 mm pitch

Pin	Signal
1	+5 V
2	TX
3	RX
4	SCL
5	SDA
6	GND

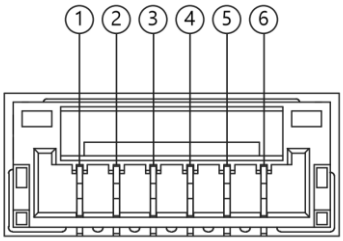
TX/RX are named from the FC perspective. Connect FC TX → device RX, FC RX ← device TX, GND ↔ GND, SCL ↔ SCL and SDA ↔ SDA. Check the peripheral power requirements.

Telemetry and I2C ports

Top view with Front at the top. Circled numbers identify pins. Left and right are reversed when viewed from the cable plug mating face.

TLM1

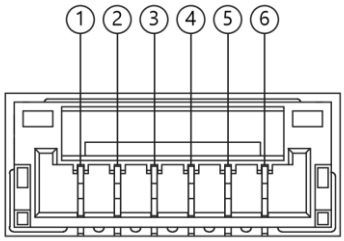
SERIAL1 / UART7



Pin	Signal
1	+5 V
2	TX
3	RX
4	CTS
5	RTS
6	GND

TLM2

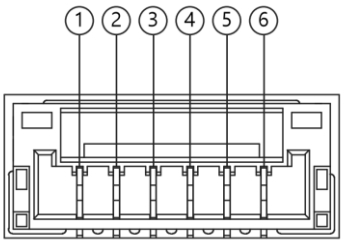
SERIAL2 / UART5



Pin	Signal
1	+5 V
2	TX
3	RX
4	CTS
5	RTS
6	GND

TLM3

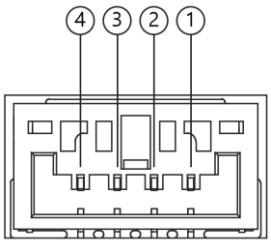
SERIAL5 / USART2



Pin	Signal
1	+5 V
2	TX
3	RX
4	SCL
5	SDA
6	GND

I2C

I2C4 / 1.25 mm pitch



Pin	Signal
1	GND
2	SDA
3	SCL
4	+5 V

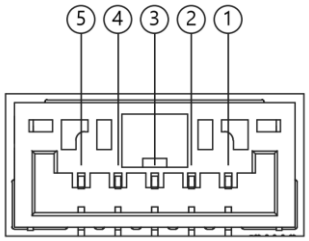
TLM1 and TLM2 have CTS/RTS. TLM3 pins 4/5 are SCL/SDA. The 5 V pin supplies peripheral power. Do not apply 5 V power to UART or I2C signal pins.

Receiver and service ports

Top view with Front at the top. Circled numbers identify pins. Left and right are reversed when viewed from the cable plug mating face.

S.B

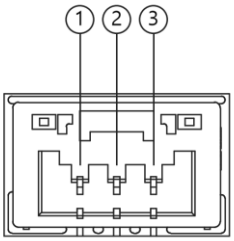
SBUS RC INPUT



Pin	Signal
1	GND
2	NC
3	RSSI
4	SBUS IN
5	+5 V

S.OUT

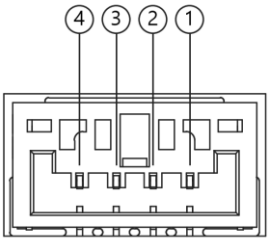
IO SBUS / OUTPUT



Pin	Signal
1	NC
2	SIGNAL OUT
3	GND

FUUP

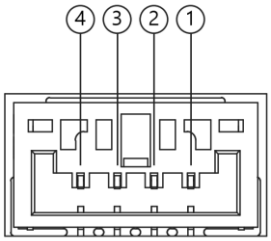
FMU SWD / SERVICE



Pin	Signal
1	3.3 V REF
2	SWCLK
3	SWDIO
4	GND

IOUP

IO SWD / SERVICE



Pin	Signal
1	3.3 V REF
2	SWCLK
3	SWDIO
4	GND

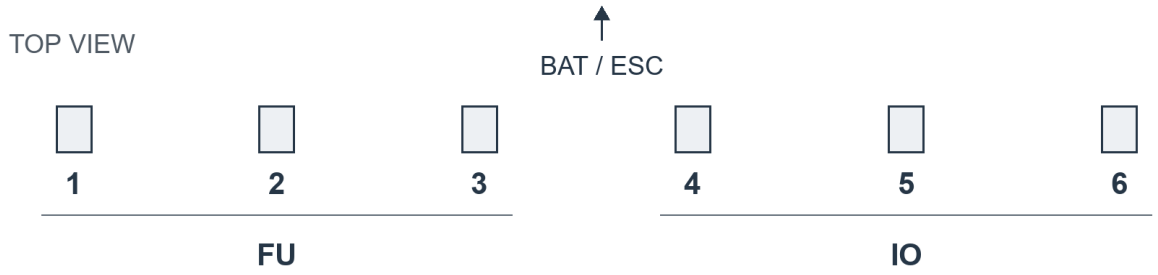
S.B is the SBUS input; S.OUT is the IO SBUS output and has no power pin. See page 10 for receiver, RSSI and SBUS output settings.

Leave NC pins unconnected. FUUP and IOUP are SWD service ports; 3.3 V REF is the debugger voltage reference.

Reset buttons and individual LEDs

FURE resets the FMU; IORE resets the IO processor. These are not DFU buttons. Do not press them during flight.

Top view of the enclosure with BAT/ESC power pads at the top. Numbers identify the LEDs in this guide.



Position	Function	Indication
1 · FU left	FMU status	Arming, pre-arm checks and calibration. See the patterns below.
2 · FU centre	GPS status	Off with no GPS or no fix. After a fix, repeated groups of flashes reflect the satellite count.
3 · FU right	FMU power	On when the FMU 3.3 V supply is present.
4 · IO left	FMU link warning	On if no FMU control data is received for over 200 ms. Off while data is being received normally.
5 · IO centre	IO steady indicator	Held on while stock IO firmware runs. Use LED 4 to check FMU communication.
6 · IO right	IO power	On when the IO 3.3 V supply is present.

LED 1 FMU status patterns

Pattern	State
Rapid flashing just after startup	Initialising
Slow single flashes while disarmed	Pre-arm checks passed
Repeated double flashes while disarmed	Pre-arm checks failed; check ground-station messages
Steady on	Armed, with no battery, RC or GCS failsafe
Flashing while armed	Battery failsafe; faster flashing for RC or GCS failsafe

LEDs 1 and 2 alternate during trim saving or ESC calibration. Both flash briefly together during compass or IMU temperature calibration.

These are the normal indications while ArduPilot runs. Bootloader, firmware-update and AutoTune notifications use other patterns. Power LEDs alone do not confirm correct operation.

UART and USB

Parameter	Connection	Default protocol
SERIAL0	First USB virtual port	MAVLink
SERIAL1	TLM1 / UART7 / CTS/RTS	MAVLink2
SERIAL2	TLM2 / UART5 / CTS/RTS	MAVLink2
SERIAL3	GPS1 / USART1	GPS
SERIAL4	GPS2 / UART8	GPS
SERIAL5	TLM3 / USART2	None
SERIAL6	USART3 / not pinned out	None
SERIAL7	Second virtual port on the same USB	MAVLink2

All UARTs have RX and TX DMA. USART6 is the internal link between the FMU and IO processors.

Loading firmware

The board ships with an ArduPilot-compatible bootloader. Connect USB and upload a DCD-FC .apj file using Mission Planner or another compatible ground station. Normal updates do not require DFU mode or ST-Link.

After board support is merged and builds are published, use the DCD-FC target at firmware.ardupilot.org.

Outputs and receiver settings

PWM and DShot

All 16 outputs support PWM and DShot. To use Bi-Directional DShot on I1-I8, set BRD_IO_DSHOT=1 and reboot.

Board outputs	ArduPilot outputs	Timer group
I1-I2	SERVO1-2	IO TIM2
I3-I4	SERVO3-4	IO TIM4
I5-I8	SERVO5-8	IO TIM3
F1-F4	SERVO9-12	FMU TIM1
F5-F6	SERVO13-14	FMU TIM4
F7-F8	SERVO15-16	FMU TIM5

Channels within the same group must use the same output rate. If any channel uses DShot, all channels in that group must use DShot.

Receiver and RSSI

Connect SBUS to S.B pin 4. PPM is not supported. Connect a CRSF/ELRS receiver to the TX/RX pins of TLM3 and set SERIAL5_PROTOCOL=23.

For analog RSSI on S.B pin 3, set RSSI_TYPE=1 and RSSI_ANA_PIN=103. S.OUT is the IO SBUS output, enabled with BRD_SBUS_OUT. SBUS output and analog RSSI cannot be used simultaneously.

GPIO

Board outputs	GPIO numbers
F1-F8	50-57
I1-I8	101-108

Use these numbers when configuring output pins as GPIOs. GPIO pins carry signals; they are not power supplies.

Compass

An IST8310 compass is fitted. Due to potential magnetic interference, an external I2C compass in a GPS/compass module is recommended.